SWIGGY INSTAMART – REGAIN MARKET POSITION

Product Analytics Report

Abstract

This report presents outcomes of a research study conducted to analyze Swiggy Instamart's current business situation & recommend performance improvements

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Executive Summary

Swiggy Instamart is fast losing its quick commerce market share to its biggest rival Blinkit and is being closely followed by a relatively new entrant Zepto. Attracting new customers who spend more on quick commerce platforms is a big challenge.

In this research study, we will address the management decision problem of improving Swiggy Instamart's market share in the quick commerce industry and elevating consumer perception about the brand by considering differentiated positioning.

The research uses qualitative & quantitative methods to analyse the business situation. The exploratory research includes studying literature, prior research work in the given context and having one-on-one discussions with quick commerce users. These qualitative methods gave valuable insights on the important factors to consider for collecting the data using survey method. Various analytical techniques such as descriptive & predictive analytics were used to analyse the data & record observations. The sample data provided enough evidence to test the research hypotheses and derive conclusions.

The research indicates that Instamart does not perform poorly in any specific area, but it does not stand out either. Its customer tenure analysis has revealed that it does not have significant number of new users. Instamart needs to spend great efforts in positioning itself with a differentiated value proposition by focusing on the insights presented by this research. It also needs to invest in communicating its unique selling proposition to the targeted customers.

Research Objectives

We have the following management decision problems to address:

1. What measures Swiggy should take to attract new customers and make the existing customers spend more on the platform?
2. How to make Swiggy stand out in the intense competition and make it attractive to other app users?
3. What opportunities Swiggy has to expand its product assortment & targeted customer segment in order to gain more market share?

The research objectives to address the management problems are:

A. How is Swiggy's performance on attributes like product selection, pricing, quality and speed compared to its competitors?
B. What factors influence customers to choose & stick to a quick commerce platform?
C. What factors contribute users to rely more on quick commerce platform for their day-to-day shopping needs?

In the research process, I have broken down the research objectives into multiple research questions and formulated the hypotheses to test using the quantitative analytics methods.

Research Questions

1. What effect overall pricing has on customer's preference for a q-commerce platform?
2. How does product quality affect customer stickiness to a q-commerce platform?
3. What effect free delivery has on customers' preference for a q-commerce app?

4. How does product selection contribute to customers loyalty to a q-commerce platform?
5. Amongst product selection, speed, quality & pricing, which factor customers value the most?
6. What effect timely/speedy delivery has on customers' satisfaction with the app?
7. On which attributes Swiggy performs better than its competitors?
8. What effect subscription model has on retaining customers and making them spend more on a q-commerce app?
9. Which product categories are not in demand on q-commerce platform and why?
10. What factors cause customers to use more than one q-commerce apps?

Research Hypotheses

1. Instamart, being Swiggy's integrated offering, has lower brand recall in the quick commerce category compared to its competitors
2. Average customer satisfaction & product selection of Instamart is lower than that of its competitors
3. Lower prices (product, delivery, handling etc.) has significant effect on customer satisfaction of a q-commerce app
4. Superior app experience & timely delivery has higher influence on customers willingness to recommend the Instamart app
5. Meeting delivery promise significantly improves customer satisfaction of the q-commerce platform
6. High quality of fresh produce like fruits, vegetables, meat significantly improves customers perception about reliability of the app

Analytics Process & Design

The following diagram depicts step wise research process design.

Figure 1: Step-wise research process

Qualitative Analytics

I studied recent research by HSBC which indicates Swiggy Instamart was leading the quick commerce market with more than half the share till 2022. However, with the entry of Blinkit and Zepto in late 2021, it gradually started losing the market share. Quick commerce consumers survey by The Ken, a digital publication, also claims that Instamart is good at all three important attributes of quick commerce – product selection, price & speed but it is exceptional at nothing. This has a consequence of not being able to attract users from competitor apps.

I also had one-on-one discussions with quick commerce users to understand different factors they consider while choosing an app, how they perceive the value offered by different platforms & decide what to shop from which app, how subscription model helps them stick to a platform and spend more. Based on the insights gathered from the qualitative methods, the research objectives & hypotheses were formulated.

From this exercise, I identified different variables for customers perception about product assortment, quality, price, speed, app reliability & user experience, return policy, availability of local brands and so on. Also identified their decision-making variables like satisfaction, likeliness of recommending the app to friends/family.
Refer the complete list of variables, including customer demographics in Appendix I.

Survey Analytics

I conducted an online survey to get the quantitative data for the variables identified in the previous step. For this survey, I have considered sample size of 116 respondents who use Swiggy Instamart and/or other quick commerce apps and have different demographics.

For this survey, I designed a structured questionnaire with simple, closed ended questions and used Likert scale to collect ratings on perception & decision-making variables and discrete scale to collect categorical data. Refer Appendix II for the survey questionnaire.

In the pilot phase, the survey questionnaire was evaluated with small sample of 10 respondents to get the feedback on whether the questions are clear, easy to understand and options cover most basic responses. The questions were rephrased wherever needed and then the survey was rolled out to the larger sample.

Data Coding

We have continuous scale values as well as discrete or categorical values in the respondent data sets. The continuous data is used as is. For discrete data, I have used dummy variables to convert all text values to numbers (0 & 1).

Descriptive Analytics

Performed descriptive analytics to get high level overview of the data such as average perception about certain variables, behavioural trends and association between different variables. I have analysed the survey data to study central tendencies such as mean, median & mode. I have also calculated range & standard deviation to understand how diverse and spread the data set is. Used correlation method to identify interdependency between different perception variables for customers.

Predictive Analytics

I created & tested various regression models to estimate the effect of different strategic actions on customer decisioning. Also, to predict their future behaviour on the basis of improvement actions that the company can take. Further, I have tested the hypotheses using regression models to infer about the population behaviour from our sample statistics.

The following regression models were used in this analytics process:

1. Estimate the effect of lower prices on customer satisfaction of the q-commerce app

2. Predict customer willing to recommend the Instamart app based on their perception about app user experience & meeting delivery promise
3. Find if the average satisfaction of Instamart users is significantly lower than that of its competitors
4. Find relationship between customer satisfaction from their perception about on time delivery
5. Find if the average product selection of Instamart is significantly lower than that of its competitors

Results

By using various analytical methods described above, I have found statistical evidence to address our research questions and test the research hypotheses.

Observations from Descriptive Analytics

Average perception about Instamart's product selection, delivery speed, app reliability & user experience is high whereas, the perception about availability of local products, quality of fresh produce & product pricing is low (Refer Fig.2)

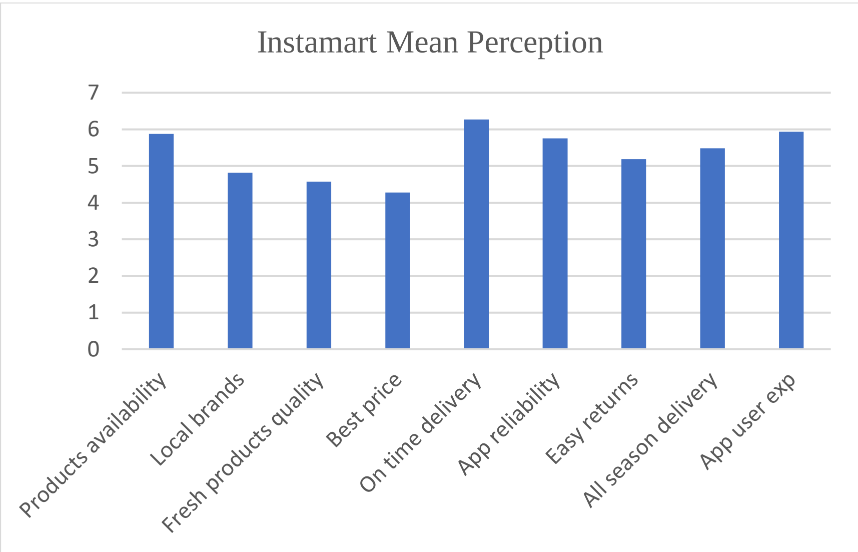


Figure 2: Mean of perception variables

The comparison between Instamart and its top competitors Blinkit & Zepto on key attributes is shown in Fig. 3. All of them fare equally on most of the attributes except for fresh products, pricing & delivery speed where Zepto scores slightly higher than the rest two.

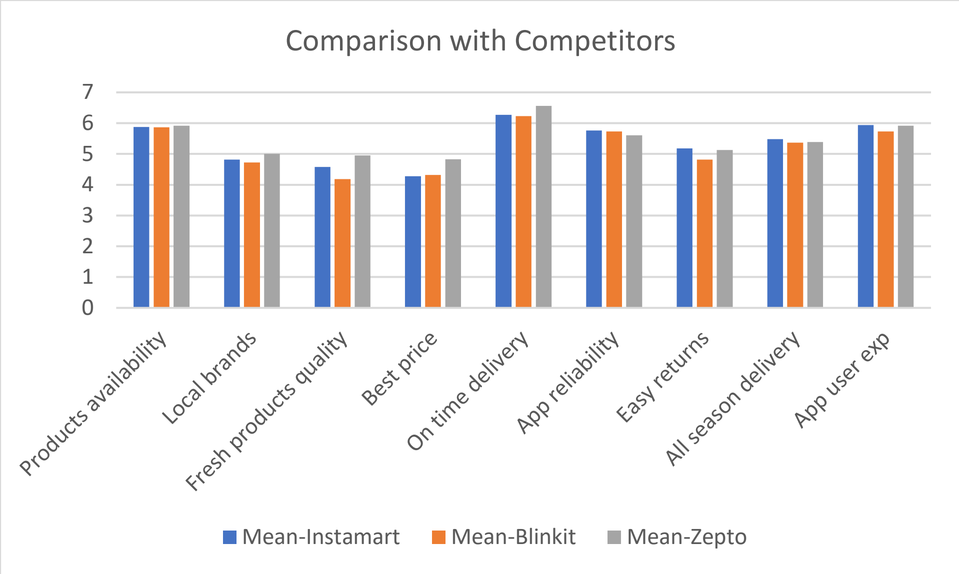


Figure 3: Comparison between Instamart & its competitors

On the app usage front, it is observed that 63% of Instamart users use the app either daily or weekly and 53% of them spend less than Rs. 500 on each order. Also, 63% respondents said less than 25% of their monthly shopping is done using the app (Refer Fig. 4). These observations are in line with the entire sample including competitors' data.

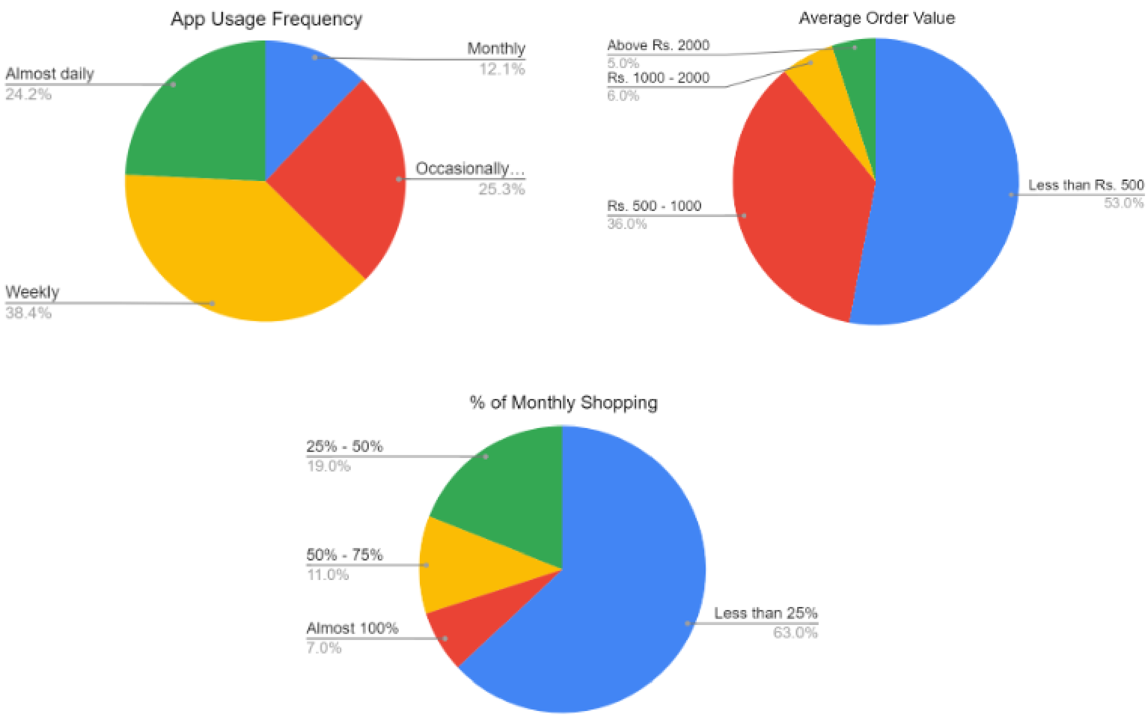
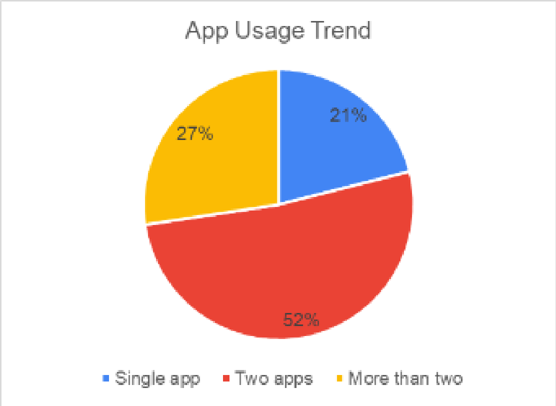


Figure 4: Instamart app usage metrics

It is also observed that quick commerce users typically use two or more apps, contributing to 80% of the total responses. The distribution of this trend over the default app suggests that users who use Blinkit and Zepto as their primary app, tend to be more habitual to quick commerce and explore more. On the other hand, Instamart (and BigBasket) users are more loyal to the platform (Refer Fig. 5).



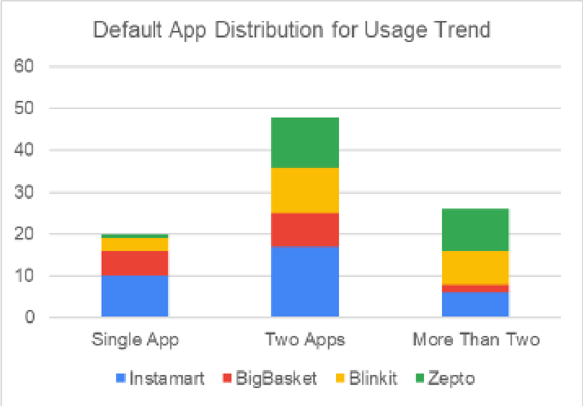


Figure 5: General app usage trend

On analysing the data for time since the users are using these apps, it is observed that Instamart has more users with 2-3 years tenure and very less since last 1 year. On the other hand, Blinkit and Zepto have considerable number of users who are using the app for 0-2 years. The sample highlights the trend that Blinkit & Zepto have attracted more new customers over the last 2 years.

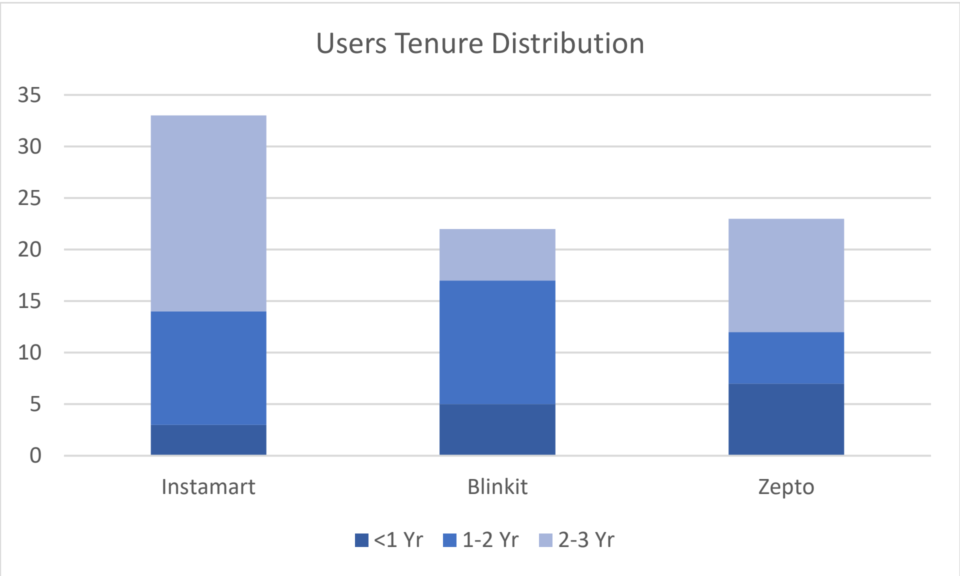


Figure 6: User tenure distribution

Key observations from correlation analysis are:

1. For Instamart, the perception about app reliability increases significantly with increase in the quality of fresh produce (correlation between the variables is 0.6)
2. In general, the perception about app reliability does not increase significantly with perception about easy returns & all-season delivery (correlation coefficient is 0.42 & 0.15 respectively which do not indicate a strong association). Users might consider them as basic must have features.

Regression Outcomes

Regression Model #1 - Estimate the effect of lower prices on customer satisfaction of the q-commerce app

Created simple regression model with customer satisfaction as dependent variable and best price as independent variable. The regression was run on the entire sample data to find out the general trend.

SUMMARY OUTPUT - Customer satisfaction

Regression Statistics	
Multiple R	0.33266562
R Square	0.11066642
Adjusted R Square	0.10110369
Standard Error	1.20001916
Observations	95

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	16.66519742	16.6652	11.5726842	0.000988279
Residual	93	133.9242763	1.440046		
Total	94	150.5894737			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	7.00250179	0.370002913	18.92553	1.1827E-33	6.267749331	7.737254243	6.267749331	7.737254243
x9_best_price	0.26594889	0.078177384	3.401865	0.00098828	0.110704101	0.421193683	0.110704101	0.421193683

R square is very less (0.11) & standard error is more than 1 which indicates the model is not at all good for prediction. The overall significance of the model is less than 5% which indicates there is significant linear relationship & the model is good for estimation

Hypothesis testing:

Null Hypothesis (H_0)	Lower prices have no significant effect on customer satisfaction of a q-commerce app
Alternate Hypothesis (H_1)	Lower prices have significant effect on customer satisfaction of a q-commerce app

The coefficient for best price is significant (P-value < 5%) indicating that the alternate hypothesis is true - Lower prices has significant effect on customer satisfaction of a q-commerce app. In statistical term it means, if the perception of best price is increased by 1 unit, customer satisfaction increases by 0.26.

Regression Model #2 – Predict customer willing to recommend the Instamart app based on their perception about app user experience & meeting delivery promise

Created multiple regression model with recommendation as dependent variable and app user experience & timely delivery as independent variables. The regression was run only on the Instamart data split from the sample.

SUMMARY OUTPUT - Willingness to recommend

Regression Statistics	
Multiple R	0.77486384
R Square	0.60041398
Adjusted R Square	0.57377491
Standard Error	1.02912816
Observations	33

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	47.7420082	23.8710041	22.53885038	1.05719E-06
Residual	30	31.77314331	1.05910478		
Total	32	79.51515152			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-2.59426589	1.570375861	-1.65200316	0.1089642	-5.80140126	0.612869476	-5.80140126	0.612869476
x15_app_user_exp	1.05236762	0.224635176	4.68478551	5.6657E-05	0.593601385	1.511133851	0.593601385	1.511133851
x10_on_time_delivery	0.67317257	0.178769711	3.76558514	0.000724107	0.30807611	1.038269023	0.30807611	1.038269023

R square is 60% & standard error is more than 1, which indicates the model is not good for prediction. The overall significance of the model is less than 5%, which indicates there is significant linear relationship and the model can be used for estimation.

Hypothesis testing:

Null Hypothesis (H_0)	Superior app experience & timely delivery have no influence on customers willingness to recommend Instamart app to friends & family
Alternate Hypothesis (H_1)	Superior app experience & timely delivery have higher influence on customers willingness to recommend Instamart app to friends & family

The coefficients for app user experience & on time delivery are significant (P-value < 5%). Hence, the sample data has enough evidence to support the alternate hypothesis - Superior app experience & timely delivery have higher influence on customers willingness to recommend the Instamart app to friends & family

Regression Model #3 - Find if the average satisfaction of Instamart users is significantly lower than that of its competitors

Conducted t-test for paired sample means to check if the mean comparison is statistically significant.

t-Test: Paired Two Sample for Means

	Customer Satisfaction - Competitors	*Customer Satisfaction - Instamart*
Mean	8.333333333	8.303030303
Variance	1.541666667	1.78030303
Observations	33	33
Pearson Correlation	0.314381198	
Hypothesized Mean Difference	0	
df	32	
t Stat	0.115278084	
P(T<=t) one-tail	0.45447253	
t Critical one-tail	1.693888748	
P(T<=t) two-tail	0.90894506	
t Critical two-tail	2.036933343	

Hypothesis testing:

Null Hypothesis (H_0)	Average satisfaction of Instamart users is same or higher than that of its competitors
Alternate Hypothesis (H_1)	Average satisfaction of Instamart users is lower than that of its competitors

The P-value is quite high (90%), which indicates the data has strong evidence to support the null hypothesis i.e. Average satisfaction of Instamart users is same or higher than that of its competitors. In this case, we reject the alternate hypothesis.

Regression Model #4 – Find relationship between customer satisfaction from their perception about on time delivery

Created simple regression model with customer satisfaction as dependent variable and on time delivery as independent variable. The regression was run on entire sample data to find out the general trend.

SUMMARY OUTPUT - Customer satisfaction

Regression Statistics	
Multiple R	0.430642067
R Square	0.18545259
Adjusted R Square	0.176694016
Standard Error	1.14845509
Observations	95

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	27.92720793	27.92720793	21.17383305	1.32382E-05
Residual	93	122.6622658	1.318949094		
Total	94	150.5894737			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	4.319846678	0.849163078	5.087181472	1.88137E-06	2.633577097	6.006116259	2.633577097	6.006116259
x10_on_time_delivery	0.613713799	0.133372455	4.601503347	1.32382E-05	0.348862535	0.878565062	0.348862535	0.878565062

R square is 18% & standard error is more than 1 which indicates the model is not at all good for prediction. The overall significance of the model is less than 5% which indicates there is significant linear relationship and the model can be used for estimation.

Hypothesis testing:

Null Hypothesis (H_0)	Meeting delivery promise has no significant effect on customer satisfaction of the q-commerce platform
Alternate Hypothesis (H_1)	Meeting delivery promise significantly improves customer satisfaction of the q-commerce platform

The coefficient for on time delivery is significant (P-value < 5%) indicating that the alternate hypothesis is true - Meeting delivery promise significantly improves customer satisfaction of the q-commerce platform. In statistical term it means, if the perception of on time delivery is increased by 1 unit, customer satisfaction increases by 0.61.

Regression Model #5 - Find if the average product selection of Instamart is significantly lower than that of its competitors

Conducted t-test for paired sample means to check if the mean comparison is statistically significant.

t-Test: Paired Two Sample for Means

	Product selection - Instamart	*Product selection - Competitors*
Mean	5.878787879	6.060606061
Variance	1.359848485	1.433712121
Observations	33	33
Pearson Correlation	0.251613392	
Hypothesized Mean Difference	0	
df	32	
t Stat	-0.722315119	
P(T<=t) one-tail	0.237673337	
t Critical one-tail	1.693888748	
P(T<=t) two-tail	0.475346674	
t Critical two-tail	2.036933343	

Hypothesis testing:

Null Hypothesis (H_0)	Average product selection of Instamart is same or greater than that of its competitors
Alternate Hypothesis (H_1)	Average product selection of Instamart users is lower than that of its competitors

Since the P value is significantly higher than 5%, the data has strong evidence to reject the alternate hypothesis. In this case, we accept the null hypothesis i.e. Average product selection of Instamart is same or greater than that of its competitors.

Limitations

The linear regression models used in this study are limited to estimation using a few variables. Models to predict future behaviour using combination of different variables is not included in this study.

While analysing regression model with multiple independent variables, we could not compare the variables since the standardised coefficient information is not available in excel. This is the limitation of using excel over other advanced analytical tools such as SPSS.

If we have time & tools like SPSS, we can perform cluster analysis to segment customers based on their satisfaction or other decision-making variables. These insights can be used to target a specific segment of customers.

Conclusions & Recommendations

Swiggy Instamart's overall performance is good compared to the industry average. It scores relatively good on the key attributes of quick commerce like product selection, speed and slightly low on pricing. Its average customer satisfaction is also in line with the industry average. Being the oldest in the industry, it has a considerable loyal customer base who exclusively shop on Instamart and have low footprint on other competitor apps.

Given Instamart is competing in an industry where competitors are equally well positioned & prepared to give a tough competition, it needs to undertake considerable strategic changes to stand out from the rest and look for the opportunities that competitors are either not considering or not good at.

The survey reveals that users do want to buy fresh produce, local products & bulk items like flour, rice, however, these products are either not listed or quality is an issue, esp. for fresh produce. The research highlights that the perception about app reliability increases significantly with increase in the quality of fresh produce. Instamart's current perception about fresh products is relatively low, which offers a great opportunity to improve in this area.

Overall, the penetration of quick commerce is still at a very early stage where over 60% of the users are only doing 25% of their monthly shopping on these apps. Also, half of them spend less than Rs. 500 on each order. There is a huge growth opportunity to increase customers' wallet share and expand product categories to increase the percentage of monthly shopping.

It is evident from the study that Instamart does not perform poorly in any specific area, but it does not stand out either. Its customer tenure analysis has revealed that it does not have significant number of new users. Instamart needs to spend great efforts in positioning itself with a differentiated value proposition by focusing on the insights presented by this research. It also needs to invest in communicating its unique selling proposition to the targeted customers.

Appendix

Appendix I: Customer variables

Variable	Type	Label	Values	Measure
x1_primary_app	Numeric	Primary or default app	0 = Instamart, 1 = Blinkit, 2 = Zepto, 3 = BigBasket, 4 = None, 5 = Other	Nominal
x2_usage_frequency	Numeric	App usage frequency	1 = Almost daily, 2 = Weekly, 3 = Monthly, 4 = Occasionally	Ordinal
x3_%monthly_spend	Numeric	Percentage of monthly shopping done on app	1 = <25%, 2 = 25-50%, 3 = 50-75%, 4 = 100%	Ordinal
x4_aov	Numeric	Average order value	1 = <500, 2 = 500-1k, 3 = 1k-2k, 4 = >2k	Ordinal
x5_most_imp_attribute	Numeric	Which attribute of q-comm business is most imp.	1 = Product availability, 2 = Speed, 3 = Pricing, 4 = Quality	Nominal
x6_product_selection	Numeric	Availability of products in size/quantity	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x7_local_brands	Numeric	Availability of local brands	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x8_product_quality	Numeric	Product quality, esp. fresh produce	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x9_pricing	Numeric	Price discounts & offers, other charges	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x10_timely_delivery	Numeric	Delivers on time as promised	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x11_reliable	Numeric	Correct order fulfilment	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x12_easy_returns	Numeric	Easy return policy, prompt customer support	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x13_order_again	Numeric	Repeat the order	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x14_all_season_delivery	Numeric	Delivery is open even in bad weather, early/late in the day	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x15_app_user_exp	Numeric	User experience of the app - ease of use	1 = Strongly Disagree, 7 = Strongly Agree	Scale
x16_satisfaction	Numeric	How satisfied using the app	1 = Not at all satisfied, 10 = Extremely satisfied	Scale
x17_highend_products	Numeric	How likely to buy highend products	1 = Very unlikely, 10 = Extremely likely	Scale
x18_recommend_friend	Numeric	How likely to recommend to friends & family	1 = Very unlikely, 10 = Extremely likely	Scale
x19_other_apps	Text	Which other q-comm apps used		Nominal
x20_reason_multiple_apps	Text	Reasons for using multiple apps		Nominal
x21_dont_buy_with reason	Text	Items don't buy and why		Free text

x22_wish_list	Text	Items wish to buy but don't find		Free text
x23_name_email	Text	Name or Email		Free text
x24_gender	Numeric	Gender		Nominal
x25_age	Numeric	Age in years		Ordinal
x26_city	Text	City live in		Free text
x27_occupation	Numeric	Occupation		Nominal
x28_customer_since	Numeric			Ordinal
x29_subscription	Text	If use subscription or membership		Nominal

Appendix II: Survey questionnaire for customers*

Survey - 10 min
Delivery Apps - Googl

Appendix III: Survey responses & results of data analytics*

Survey Responses &
Data Analysis.xlsx

**Please note that the documents attached here are also uploaded on the Canvas portal.*